

United States using 110v and we ourselves use 230v. Earlier power supplies had a small switch in the back to select the voltage, but with improved technology this is accomplished within the SMPS itself.

Why switching?

Power supplies in general can be either Linear (only uses a transformer to bring down AC power supply before rectifying it into DC) or it can be switching as most power supplies are today. Now the difference between 230v AC and 12v DC is pretty huge, so the transformer used to convert this would traditionally be huge as is the case in linear power supplies. Now the number of coils of winding in the transformer is dependent on the frequency of the input voltage, the lower the frequency the bigger the transformer. So switching mode power supplies increase the frequency of the input voltage, thus, reducing the size of the transformer and subsequently the size of the power supply unit.

Form factors

Now computer cabinets come in all shapes and sizes given the way it is to be implemented, server power supplies are lean and long while desktop power supplies look like one half of a brick. When computers are concerned there are two form factors which are the most popular, ATX12V and EPS12V. The standards not only dictate the dimensions of the outer box but also the values of voltages that are in the power supply.

ATX

A standard throughout the 90s and the early 2000s, the ATX form factor is what standardised by IBM. It is difficult to get your hands on any of these now but they were notable in bringing about a few changes. The 3.3v rail was introduced with this form factor and so was the 20-pin motherboard connector. The major change was that you could now use your operating system to power down the system. Back in the days of Windows 95, your PC would not power down completely, rather it would reach a state where you had to manually switch it off. We now have the revised version of ATX where the 20-pin motherboard connector is replaced with a 24-pin connector and certain changes were made as to how much current should be allowed on each rail. CPUs were given separate connectors and no longer shared the motherboard's power supply.